

CONTACT INFORMATION	Oxford, MS 38655 gavin.s.davies@gmail.com · gsdavies@olemiss.edu	(+1) 331-442-3845 https://gavinsdavies.com
BACKGROUND	Experimental High Energy physicist studying neutrino oscillations, with leadership roles in the software and computing infrastructure underpinning large-scale international collaborations	
PROFESSIONAL EXPERIENCE	Associate Professor of Physics & Astronomy University of Mississippi , Dept. of Physics & Astronomy, Oxford, MS USA University of Mississippi Center for Multi-Messenger Astrophysics Affiliate	July 2025 - Present
	Assistant Professor of Physics & Astronomy University of Mississippi , Dept. of Physics & Astronomy, Oxford, MS USA University of Mississippi Center for Multi-Messenger Astrophysics Affiliate	Aug. 2019 - July 2025
	Postdoctoral Research Fellow Indiana University , Dept. of Physics, Bloomington, IN USA Supervisor: Prof. Mark Messier	Oct. 2014 - Aug. 2019
	Postdoctoral Researcher Iowa State University , Dept. of Physics, Ames, IA USA Supervisor: Prof. Mayly Sanchez	Jul. 2010 - Oct. 2014
	Doctor of Philosophy (Ph.D.) in Physics Lancaster University , Dept. of Physics, Lancaster, UK Dissertation: 'Understanding and calibrating the DS-ECal of ND280 in the T2K project' Ph.D. Advisor: Dr. Laura Kormos	Oct. 2006 - July 2010
EDUCATION	M.Phys (Hons.) Physics / USA Lancaster University , Dept. of Physics, Lancaster, UK Dissertation: 'A Study of Charge Transfer Inefficiency in a Simulated CCD Damaged by Particle Radiation'	Oct. 2002 - July 2006
	USA Study Abroad Program, Michigan State University , MI USA	Aug. 2004 - May 2005
LANGUAGES	English (native) · Spanish (B1) · French (A2) · Danish (A1)	
LEADERSHIP EXPERIENCE	DUNE Phlex Adoption Working Group co-lead	May. 2026 - Present
	DUNE Global Software Technical Lead	Jan. 2026 - Present
	EMPHATIC Software & Analysis Coordinator	Oct. 2022 - Present
	NOvA Institutional Board Chairperson Elected position	Mar. 2021 - Mar. 2023
	NOvA Computing Coordinator	Jan. 2020 - Present
	Fermilab Users Executive Committee Chairperson Elected position	Sep. 2018 - Sep. 2019

	NOvA Exotic Neutrino Oscillation Analysis Convener	Oct. 2017 - Sep. 2022
	NOvA Production convener, Software Framework manager	Feb. 2011 - July 2014
COMMITTEES, SERVICE, AND AWARDS	Journal of Open Source Software (JOSS) Peer Reviewer	June 2026 - Present
	APS DPF Coordinating Panel for Software & Computing [CPSC] Panel member, Rising Star Awards Sub-Committee Lead	Mar. 2025 - Present
	UM Excellence in Community Engaged Teaching and Learning Award Team member, Dept. of Physics & Astronomy Astronomy Art Contest	2024, 2025
	International Conference of High Energy Physics [ICHEP24] Computing co-chair, Prague	July 2024
	Graduate Program Coordinator, Dept. of Physics & Astronomy, University of Mississippi	Jan. 2025 - Present
	APS Formation Task Force for the Coordinating Panel for Software and Computing	Dec. 2023 - July 2024
	APS DPF Particle Physics Planning: Topical Group convener End-User Analysis	Apr. 2020 - July 2023
	Department of Energy Office of Science User Facilities Roundtable Committee member	Oct. 2020 - June 2021
	Flavor Physics and CP Violation (FPCP) international conference Organizing Committee	May 2021 - May 2022
	NOvA Executive Committee Elected position	May 2018 - Present
	Advocacy for High-Energy Physics Funding UEC Washington D.C. Trip	2017 - Present
	Fermilab Users Executive Committee (UEC) Elected position	Sep. 2017 - Sep. 2019
	Breakthrough Prize in Fundamental Physics	2016
AFFILIATIONS	APS DPF · SESAPS · NOvA Collaboration · EMPHATIC Collaboration · DUNE Collaboration	
OUTREACH & PUBLIC ENGAGEMENT	Dept. of Physics & Astronomy Astronomy Art Contest Judge	2024, 2025

Mississippi Region VII Science Fair Judge	2025
University of Mississippi Academic Quiz Tournament Judge	2020, 2021, 2023, 2024
8th Annual Mississippi FIRST Tech Challenge Robotics Championship Judge	Feb. 2020
UM Spooky Physics Annual Halloween participant (Mr. Freeze demo)	2022 - Present
Oxford Science Cafe Quizmaster (during COVID-19 pandemic)	2020
Graduate Fair Informational Tables Dept. of Physics & Astronomy representative	various, 2021 - Present

GRANTS

External Funding; Total Awarded: \$4,576,992

Establishing a Scientific Computing Hub at the University of Mississippi Principal Investigator, U.S. Dept. of Energy (EPSCoR), Award DE-SC0026215, \$1,800,000 Co-PIs: B. Quinn, J. Bennett (U. Mississippi), S. Tu, S. Yang (Jackson State U.)	Sep. 2025 – Aug. 2027
LBNF/DUNE-US Operations Principal Investigator, Fermilab, Award #734754, \$13,349	Apr. 2026 – Mar. 2027
URA Visiting Scholars Program: EMPHATICally Preparing for DUNE Principal Investigator, Universities Research Association, Award #2026-S-24, \$17,000	Jun. 2026 – Nov. 2026
URA Visiting Scholars Program: EMPHATICally PINning DUNE Principal Investigator, Universities Research Association, Award #2025-S-13, \$16,000	Jun. 2025 – May 2026
LBNF/DUNE-US Operations Principal Investigator, Fermilab, \$27,143	Oct. 2025 – Mar. 2026
Intensity Frontier Studies at the University of Mississippi co-Principal Investigator, U.S. Dept. of Energy (HEP), \$1,895,000 Co-PIs: B. Quinn, J. Bennett (U. Mississippi)	Sep. 2023 – Mar. 2026
Intensity Frontier Studies at the University of Mississippi co-Principal Investigator, U.S. Dept. of Energy (HEP), Award DE-SC0021616, \$800,000 Co-PI: B. Quinn (U. Mississippi)	Nov. 2020 – Apr. 2023
Stimulating and Supporting Interest in STEM Fields with Hands-On Masterclass Programs co-Principal Investigator, University of Mississippi Mississippi Impact Grant, Account 250223370A,	May 2025 – Dec. 2026

\$8,500
PI: J. Bennett (U. Mississippi)

Travel Awards

ORSP Travel Grant May 2026
Travel to CHEP 2026 (Bangkok, Thailand), parallel talk, U. Mississippi ORED, \$1,200

ORSP Travel Grant Oct. 2024
Travel to NBI 2024 (Mito, Japan), invited talk, U. Mississippi ORED, \$1,200

ORSP Travel Grant Jul. 2024
Travel to ICHEP 2024 (Prague, Czechia), computing track convenor, U. Mississippi ORED, \$1,188

Pending / Under Review; Total Pending: \$4,162,950

Intensity Frontier Studies at the University of Mississippi Apr. 2026 – Mar. 2029
co-Principal Investigator, U.S. Dept. of Energy (HEP), \$3,076,482 requested
Co-PIs: B. Quinn, J. Bennett (U. Mississippi)

FEC: From Fundamental Physics to Real-World Applications: Training the Next Generation in AI/ML and Image Recognition Sep. 2026 – Aug. 2030
co-Principal Investigator, NSF EPSCoR (Research Infrastructure Improvement: Focused EPSCoR Collaborations), \$861,066 (UM share of \$5,865,499 total)
Multi-jurisdiction proposal: Wichita State U. (lead), U. Kansas, Kansas State U., U. Nebraska, U. Mississippi

AI-Enhanced DUNE Framework and Simulation Performance Jul. 2026 – Mar. 2027
Collaborator, DOE Genesis Mission subaward (PNNL lead), \$112,701 (UM share of \$750,000 total)

Phlex as a Trustworthy AI Framework Jul. 2026 – Mar. 2027
Collaborator, DOE Genesis Mission subaward (Fermilab lead), \$112,701 (UM share of \$750,000 total)

TEACHING/STUDENT SUPERVISION

Teaching

- PHYS 211, 212 Physics for Scientists and Engineers I/II
- PHYS 709 Advanced Mechanics (graduate-level)
- PHYS 503 Science Communication (graduate-level)
- PHYS 510 Department Colloquium

Postdoctoral Researchers

- Daniel Souza Correia, University of Mississippi 2026 - Present
DUNE
- Jeffrey Kleykamp, University of Mississippi 2021 - Present
NOvA / DUNE

Ph.D. Students

- Noah Knutson, University of Mississippi 2024 - Present
EMPHATIC / DUNE
- Xiaoyan Huang, University of Mississippi 2023 - Present
DUNE / NOvA

- Bishnu Acharya, University of Mississippi 2021 - Present
NOvA
- Andrew Dye, University of Mississippi 2021 - May 2025
Ph.D. Physics (NOvA)
Now postdoctoral researcher at Wichita State University
- Luiz Ricardo Prais, University of Mississippi 2021 - Nov. 2024
Ph.D. Physics (NOvA)
Now postdoctoral researcher at University of Cincinnati

M.Sc. Students

- Ehsan Farooq, University of Mississippi 2024 - July 2025
M.Sc. Physics (DUNE)
Now pursuing Biomedical Engineering Ph.D. at University of Mississippi
- Devesh Bhattarai, University of Mississippi 2020 - Aug. 2023
M.Sc. Physics (NOvA / DUNE)
Now pursuing Physics Ph.D. at Mississippi State University

Undergraduate Students

- Suyog Badal, University of Mississippi Summer 2025 - Present
NOvA / DUNE, undergraduate research
- Samantha Reyes, University of Mississippi Summer 2026 - Present
MicroBooNE / DUNE machine learning, undergraduate research
- Chad Byrd, University of Mississippi Summer 2026 - Present
MicroBooNE / DUNE machine learning, undergraduate research
- Kaylyn Beard, University of Mississippi Summer 23, Fall 24
NOvA, undergraduate research
Now pursuing Physics Ph.D. at University of Minnesota

Additional Thesis Committee Service

- Served on 21 additional undergraduate honors, M.Sc., and Ph.D. thesis and prospectus committees since 2019, spanning Belle II, LIGO, Muon g-2, and theory in addition to NOvA/DUNE

EPIC (Exploring particle Physics Integrated with Computing) Summer School

- Coordinator and mentor, undergraduate summer training programme Jan. 2026 - Present
Mississippi EPSCoR Scientific Hub (MESH); recruitment, curriculum development, and student mentorship

Pedagogical Development

- ARISE@UM mentor
- CETL workshop, 'Energizing and Motivating Students to Learn in Uncertain and Challenging Times'
- CETL workshop, 'Top 5 Teaching Tips from Recent Pedagogical Research'

SELECTED PRESENTATIONS

June 2026 'New directions for joint neutrino oscillation measurements with T2K and NOvA' (poster)
Neutrino 2026, University of California, Irvine, CA

June 2026 'Probing Non-Standard Interactions in NOvA with Bayesian MCMC methods' (poster)
Neutrino 2026, University of California, Irvine, CA

May 2026 'NOvA Software and Computing: Evolution, Lessons, and Impact; Supporting a Decade of Physics' (contributed talk)
CHEP 2026, Bangkok, Thailand

Apr. 2026 ‘Ten Years of NOvA in the Current Long-Baseline Picture, with DUNE on the Horizon’ (invited seminar)

Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark

Dec. 2024 ‘Exploring Exotic Neutrino Oscillations with the NOvA Experiment’ (invited plenary talk)

Miami 2024, University of Miami, Fort Lauderdale, FL

Nov. 2024 ‘Long-baseline neutrino experiments: Today and tomorrow’ (invited parallel talk)

SESAPS 2024, UNC Charlotte, North Carolina

Oct. 2024 ‘EMPHATIC’ (invited plenary talk)

NBI 2024, Neutrino Beams and Instrumentation, Tokai, Japan

Oct. 2024 ‘Hadron production measurements at EMPHATIC’ (invited parallel talk)

J-PARC symposium, Mito, Japan

Dec. 2023 ‘Long-baseline neutrino experiments: hoy y mañana’ (invited plenary talk)

8th ComHEP: Colombian Meeting on High Energy Physics, Ibagué, Colombia

July 2022 ‘Measurement of Standard and Non-standard Oscillations at NOvA’ (invited plenary talk)

XLI International Conference on High Energy Physics [ICHEP 2022], Bologna, Italy

Nov. 2021 ‘Latest oscillation results from NOvA’ (invited parallel talk)

SESAPS 2021, Tallahassee, Florida

Nov. 2019 ‘Latest three-flavor neutrino oscillation results from NOvA’ (invited parallel talk)

SESAPS 2019, Wrightsville Beach, North Carolina

Nov. 2019 ‘The Fermilab Users Executive Committee Model’ (invited parallel talk)

Society for Science at User Research Facilities (SSURF) annual meeting, College Park

Oct. 2019 ‘NOvA Software Infrastructure’ (invited parallel talk)

Inaugural Dark-matter and Neutrino Computation Explored (DANCE) workshop, Houston, TX

Aug. 2019 ‘Latest three-flavor neutrino oscillation results from NOvA’ (invited parallel talk)

Lepton Photon 2019, Toronto, Canada

Mar. 2019 ‘NOvA and T2K: Status and Outlook’ (invited plenary talk)

Aspen Winter Conference, Aspen, Colorado

Aug. 2018 ‘Neutrinos, antineutrinos, and deep learning on NOvA’ (invited seminar)

G.S. Davies, Seminar talk at Wichita State University, Wichita, KS

Sep. 2017 ‘Results and Prospects from NOvA’ (invited plenary talk)

19th International Workshop on Neutrinos from Accelerators (NuFACT 2017), Uppsala, Sweden

Aug. 2016 ‘Searches for Sterile Neutrinos with NOvA’ (invited parallel talk)

38th International Conference on High Energy Physics (ICHEP 2016), Chicago, Illinois, USA

Aug. 2016 ‘First Results from Searches for Active to Sterile Neutrino Oscillations with NOvA’
Fermilab JETP (Wine & Cheese) Seminar

SOFTWARE

nuosclab (Neutrino Oscillation Laboratory) 2026 – Present
Interactive PMNS + NSI neutrino oscillation probability viewer, with a frontend-neutral computation API and pluggable, cross-validated oscillation engines. v1.0.0, DOI: [10.5281/zenodo.21919616](https://doi.org/10.5281/zenodo.21919616).
github.com/gavinsdavies/nuosclab

SELECT PUBLICATIONS

115 refereed and non-refereed publications, 13,090+ citations, h-index 40 (INSPIRE-HEP, July 2026; citation/h-index not re-pulled since)

Recent Refereed Publications (2024-2026)

- S. Abbaslu et al. (DUNE Collaboration), “Reconstruction of atmospheric neutrinos in DUNE’s horizontal-drift far-detector module”, *Eur. Phys. J. C* 86 7, 904 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Measurement of π^0 Production in $\bar{\nu}_\mu$ Charged-Current Interactions in the NOvA Near Detector”, *Phys. Rev. D* 113, 112008 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Ionization-based search for magnetic monopoles using the NOvA Far Detector”, *Phys. Rev. D* 113, 112014 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Charge readout electronics for the DUNE horizontal drift far detector: design and performance in ProtoDUNE-HD”, *JINST* 21 08, P08018 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Identification of low-energy kaons in the ProtoDUNE-SP detector”, *Phys. Rev. D* 113, 052004 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Operation of a Modular 3D-Pixelated Liquid Argon Time-Projection Chamber in a Neutrino Beam”, *Instruments* 10, 18 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Precision Measurement of Neutrino Oscillation Parameters with 10 Years of Data from the NOvA Experiment”, *Phys. Rev. Lett.* 136 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Explanation of the seasonal variation of cosmic multiple muon events observed with the NOvA Near Detector”, *Phys. Rev. D* 113 (2026)
- M. A. Acero et al. (NOvA Collaboration), “Measurement of $d^2\sigma/d|q|dE_{avail}$ in charged current neutrino-nucleus interactions at $\langle E_\nu \rangle = 1.86$ GeV using the NOvA Near Detector”, *Phys. Rev. D* 111, 052009 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations in NOvA”, *Phys. Rev. Lett.* 134, 081804 (2025)
- M. A. Acero et al. (NOvA and T2K Collaborations), “Joint neutrino oscillation analysis from the T2K and NOvA experiments”, *Nature* 646, 818 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Monte Carlo method for constructing confidence intervals with unconstrained and constrained nuisance parameters in the NOvA experiment”, *JINST* 20 02, T02001 (2025)
- S. Abbaslu et al. (DUNE Collaboration), “Spatial and temporal evaluations of the liquid argon purity in ProtoDUNE-SP”, *JINST* 20 09, P09008 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The track-length extension fitting algorithm for energy measurement of interacting particles in liquid argon TPCs and its performance with ProtoDUNE-SP data”, *JINST* 20 02, P02021 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “Supernova pointing capabilities of DUNE”, *Phys. Rev. D* 111 9, 092006 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Search for CP-violating Neutrino Non-Standard Interactions with the NOvA Experiment”, *Phys. Rev. Lett.* 133 20, 201802 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “First Measurement of the Total Inelastic Cross-

Section of Positively-Charged Kaons on Argon at Energies Between 5.0 and 7.5 GeV”, Phys. Rev. D 110 9, 092011 (2024)

- M. A. Acero et al. (NOvA Collaboration), “Expanding neutrino oscillation parameter measurements in NOvA using a Bayesian approach”, Phys. Rev. D 110, 012005 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Far Detector Vertical Drift Technology. Technical Design Report”, JINST 19 08, T08004 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “DUNE Phase II: scientific opportunities, detector concepts, technological solutions”, JINST 19 12, P12005 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “Performance of a Modular Ton-Scale Pixel-Readout Liquid Argon Time Projection Chamber”, Instruments 8 3, 41 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “Doping liquid argon with xenon in ProtoDUNE Single-Phase: effects on scintillation light”, JINST 19 08, P08005 (2024)

Publications (2020-2023)

- M. A. Acero et al. (NOvA Collaboration), “Measurement of the ν_e -Nucleus Charged-Current Double-Differential Cross Section at $\langle E_\nu \rangle = 2.4$ GeV using NOvA”, Phys. Rev. Lett. 130 (2023) 5, 051802 (2023)
- A. Abed Abud et al. (DUNE Collaboration), “Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino Experiment”, Phys. Rev. D 107, 112012 (2023)
- M. A. Acero et al. (NOvA Collaboration), “Improved measurement of neutrino oscillation parameters by the NOvA experiment”, Phys. Rev. D 106, 032004 (2022)
- A. Abed Abud et al. (DUNE Collaboration), “Design, construction and operation of the ProtoDUNE-SP Liquid Argon TPC”, JINST 17 01, P01005 (2022)
- M. A. Acero et al. (NOvA Collaboration), “Search for Active-Sterile Antineutrino Mixing Using Neutral-Current Interactions with the NOvA Experiment”, Phys. Rev. Lett. 127, 201801 (2021)
- B. Abi et al. (DUNE Collaboration), “Supernova neutrino burst detection with the Deep Underground Neutrino Experiment”, Eur. Phys. J. C 81 5, 423 (2021)
- B. Abi et al. (DUNE Collaboration), “First results on ProtoDUNE-SP liquid argon time projection chamber performance from a beam test at the CERN Neutrino Platform”, JINST 15 12, P12004 (2020)
- B. Abi et al. (DUNE Collaboration), “Long-baseline neutrino oscillation physics potential of the DUNE experiment”, Eur. Phys. J. C 80 10, 978 (2020)
- B. Abi et al. (DUNE Collaboration), “Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume IV: Far Detector Single-phase Technology”, JINST 15 08, T08010 (2020)

Publications (2016-2019)

- M. A. Acero et al. (NOvA Collaboration), “First Measurement of Neutrino Oscillation Parameters using Neutrinos and Antineutrinos by NOvA”, Phys. Rev. Lett. 123 151803 (2019)
- M. A. Acero et al. (NOvA Collaboration), “New constraints on oscillation parameters from ν_e appearance and ν_μ disappearance in the NOvA experiment”, Phys. Rev. D 98, 032012 (2018)
- P. Adamson et al. (NOvA Collaboration), “Search for active-sterile neutrino mixing using neutral-current interactions in NOvA”, Phys. Rev. D. 96, 072006 (2017)
- P. Adamson et al. (NOvA Collaboration), “Constraints on Oscillation Parameters from ν_e Appearance and ν_μ Disappearance in NOvA”, Phys. Rev. Lett. 118, 231801 (2017)
- P. Adamson et al. (NOvA Collaboration), “Measurement of the neutrino mixing angle θ_{23} in NOvA”, Phys. Rev. Lett. 118, 151802 (2017)
- P. Adamson et al. (NOvA Collaboration), “First Measurement of Electron Neutrino Appearance in NOvA”, Phys. Rev. Lett. 116, 151806 (2016)
- P. Adamson et al. (NOvA Collaboration), “First Measurement of muon-neutrino disappearance in NOvA”, Phys. Rev. D 93, 051104(R) (2016)

T2K Collaboration

- D. Allan et al. (T2K Collaboration), “The Electromagnetic Calorimeter for the T2K Near Detector”, JINST 8 P10019 (2013)
- K. Abe et al. (T2K Collaboration), “First Muon-Neutrino Disappearance Study with an Off-Axis Beam”, Phys. Rev. D. 85, 031103 (2012)
- K. Abe et al. (T2K Collaboration), “Indication of Electron Neutrino Appearance from an Accelerator-produced Off-axis Muon Neutrino Beam”, Phys. Rev. Lett. 107, 041801 (2011)

Selected Non-refereed Publications

- S. Abubakar et al. (NOvA Collaboration), “The NOvA Test Beam Experiment”, arXiv:2609.10804 (2026)
- S. Abubakar et al. (NOvA Collaboration), “NOvA Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations using Neutrino- and Antineutrino-Enriched Samples”, arXiv:2609.08900 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Neutron detector response modeling in NOvA”, arXiv:2609.06956 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Measurement of the $\bar{\nu}_\mu$ -Hydrogen Charged-Current Quasi-Elastic Cross Section using the NOvA Near Detector”, arXiv:2608.12293 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “A Bayesian approach to the long-baseline neutrino oscillation sensitivity of DUNE”, arXiv:2608.04059 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Timing-Based Search for Magnetic Monopoles with the NOvA Detector on the Surface”, arXiv:2608.03888 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Operation and performance of ProtoDUNE Dual Phase liquid argon time projection chamber”, arXiv:2607.15927 (2026)
- S. Abubakar et al. (NOvA Collaboration), “Determining Neutrino Mass Ordering with NOvA and Upcoming JUNO Measurements”, arXiv:2606.14121 (2026)
- S. Abubakar et al. (NOvA Collaboration), “First inclusive triple-differential measurement of the muon-antineutrino charged-current cross section using the NOvA Near Detector”, arXiv:2603.06718 (2026)
- T. K. Aarrestad et al. (HEP Community), “Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision”, arXiv:2602.17582 (2026)
- S. Abbaslu et al. (DUNE Collaboration), “Measurement of Exclusive π^+ -argon Interactions Using ProtoDUNE-SP”, arXiv:2511.13462 (2025)
- S. Abbaslu et al. (DUNE Collaboration), “First Measurement of π^+ -Ar and p -Ar Total Inelastic Cross Sections in the Sub-GeV Energy Regime with ProtoDUNE-SP Data”, arXiv:2511.11925 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “European Contributions to Fermilab Accelerator Upgrades and Facilities for the DUNE Experiment”, arXiv:2503.23744 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “DUNE Software and Computing Research and Development”, arXiv:2503.23743 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Science Program”, arXiv:2503.23291 (2025)
- A. Abed Abud et al. (DUNE Collaboration), “The DUNE Phase II Detectors”, arXiv:2503.23293 (2025)
- M. A. Acero et al. (NOvA Collaboration), “Measurement of the double-differential cross section of muon-neutrino charged-current interactions with low hadronic energy in the NOvA Near Detector”, arXiv:2410.10222 (2024)
- A. Abed Abud et al. (DUNE Collaboration), “DUNE Offline Computing Conceptual Design Report”, arXiv:2210.15665 (2022)
- V. Daniel Elvira et al., “The Future of High Energy Physics Software and Computing”, Snowmass 2021, arXiv:2210.05822 (2022)
- G.S. Davies, P. Onyisi, and A. Roberts, “CompF5: End User Analysis Topical Group Report”, Snowmass 2021, arXiv:2209.14984 (2022)
- A. Abed Abud et al. (DUNE Collaboration), “Snowmass Neutrino Frontier: DUNE Physics Summary”, Snowmass 2021, arXiv:2203.06100 (2022)

- B. Abi et al. (DUNE Collaboration), “Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume II: DUNE Physics”, arXiv:2002.03005 (2020)

Full INSPIRE Publication List

- The NOvA Test Beam Experiment, arXiv:2609.10804 (2026)
- NOvA Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations using Neutrino- and Antineutrino-Enriched Samples, arXiv:2609.08900 (2026)
- Neutron detector response modeling in NOvA, arXiv:2609.06956 (2026)
- Measurement of the $\bar{\nu}_\mu$ -Hydrogen Charged-Current Quasi-Elastic Cross Section using the NOvA Near Detector, arXiv:2608.12293 (2026)
- A Bayesian approach to the long-baseline neutrino oscillation sensitivity of DUNE, arXiv:2608.04059 (2026)
- Timing-Based Search for Magnetic Monopoles with the NOvA Detector on the Surface, arXiv:2608.03888 (2026)
- Operation and performance of ProtoDUNE Dual Phase liquid argon time projection chamber, arXiv:2607.15927 (2026)
- Determining Neutrino Mass Ordering with NOvA and Upcoming JUNO Measurements, arXiv:2606.14121 (2026)
- Design of the DUNE horizontal drift far detector charge readout electronics and performance in ProtoDUNE-HD, arXiv:2604.23966 (2026)
- First inclusive triple-differential measurement of the muon-antineutrino charged-current cross section using the NOvA Near Detector, arXiv:2603.06718 (2026)
- Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision, arXiv:2602.17582 (2026)
- Reconstruction of atmospheric neutrinos in DUNE's horizontal-drift far-detector module, arXiv:2601.05697 (2026)
- Ionization-based search for magnetic monopoles using the NOvA Far Detector, Phys. Rev. D 113, 112014 (2026), arXiv:2512.20294
- Measurement of Exclusive $+$ $-$ argon Interactions Using ProtoDUNE-SP, arXiv:2511.13462 (2025)
- First Measurement of $+$ $-$ Ar and p $-$ Ar Total Inelastic Cross Sections in the Sub-GeV Energy Regime with ProtoDUNE-SP Data, arXiv:2511.11925 (2025)
- Measurement of 0 Production in $\bar{\nu}_\mu$ Charged-Current Interactions in the NOvA Near Detector, arXiv:2511.05807 (2025)
- Joint neutrino oscillation analysis from the T2K and NOvA experiments, arXiv:2510.19888 (2025)
- Identification of low-energy kaons in the ProtoDUNE-SP detector, arXiv:2510.08380 (2026)
- Operation of a Modular 3D-Pixelated Liquid Argon Time-Projection Chamber in a Neutrino Beam, arXiv:2509.07012 (2026)
- Precision Measurement of Neutrino Oscillation Parameters with 10 Years of Data from the NOvA Experiment, arXiv:2509.04361 (2026)
- Explanation of the seasonal variation of cosmic multiple muon events observed with the NOvA Near Detector, arXiv:2508.04434 (2026)
- Spatial and temporal evaluations of the liquid argon purity in ProtoDUNE-SP, arXiv:2507.08586 (2025)
- European Contributions to Fermilab Accelerator Upgrades and Facilities for the DUNE Experiment, arXiv:2503.23744 (2025)
- DUNE Software and Computing Research and Development, arXiv:2503.23743 (2025)
- The DUNE Science Program, arXiv:2503.23291 (2025)
- The DUNE Phase II Detectors, arXiv:2503.23293 (2025)
- Measurement of the double-differential cross section of muon-neutrino charged-current interactions with low hadronic energy in the NOvA Near Detector, arXiv:2410.10222 (2024)
- Measurement of $d^2/d|q|dE_{\text{avail}}$ in charged current $-$ nucleus interactions at $E = 1.86$ GeV using the NOvA Near Detector, arXiv:2410.05526 (2025)
- The track-length extension fitting algorithm for energy measurement of interacting particles in liquid argon TPCs and its performance with ProtoDUNE-SP data, arXiv:2409.18288 (2025)
- Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations in NOvA, arXiv:2409.04553 (2025)
- DUNE Phase II: scientific opportunities, detector concepts, technological solutions, arXiv:2408.12725 (2024)

- First measurement of the total inelastic cross section of positively charged kaons on argon at energies between 5.0 and 7.5 GeV, arXiv:2408.00582 (2024)
- Supernova pointing capabilities of DUNE, arXiv:2407.10339 (2025)
- Search for CP-Violating Neutrino Nonstandard Interactions with the NOvA Experiment, arXiv:2403.07266 (2024)
- Performance of a Modular Ton-Scale Pixel-Readout Liquid Argon Time Projection Chamber, arXiv:2403.03212 (2024)
- Doping liquid argon with xenon in ProtoDUNE Single-Phase: effects on scintillation light, arXiv:2402.01568 (2024)
- The DUNE Far Detector Vertical Drift Technology. Technical Design Report, arXiv:2312.03130 (2024)
- Expanding neutrino oscillation parameter measurements in NOvA using a Bayesian approach, arXiv:2311.07835 (2024)
- Measurement of charged-current inclusive Λ production in the NOvA near detector, arXiv:2306.04028 (2023)
- Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino Experiment, arXiv:2303.17007 (2023)
- Highly-parallelized simulation of a pixelated LArTPC on a GPU, arXiv:2212.09807 (2023)
- Identification and reconstruction of low-energy electrons in the ProtoDUNE-SP detector, arXiv:2211.01166 (2023)
- DUNE Offline Computing Conceptual Design Report, arXiv:2210.15665 (2022)
- CompF5: End User Analysis Topical Group Report, arXiv:2209.14984 (2022)
- Monte Carlo method for constructing confidence intervals with unconstrained and constrained nuisance parameters in the NOvA experiment, arXiv:2207.14353 (2025)
- Reconstruction of interactions in the ProtoDUNE-SP detector with Pandora, arXiv:2206.14521 (2023)
- Measurement of the νe^- Nucleus Charged-Current Double-Differential Cross Section at $\sqrt{s} = 2.4$ GeV using NOvA, arXiv:2206.10585 (2023)
- Separation of track- and shower-like energy deposits in ProtoDUNE-SP using a convolutional neural network, arXiv:2203.17053 (2022)
- Scintillation light detection in the 6-m drift-length ProtoDUNE Dual Phase liquid argon TPC, arXiv:2203.16134 (2022)
- Dark-matter And Neutrino Computation Explored (DANCE) Community Input to Snowmass, arXiv:2203.08338 (2022)
- White paper on light sterile neutrino searches and related phenomenology, arXiv:2203.07323 (2024)
- Snowmass Neutrino Frontier: DUNE Physics Summary, arXiv:2203.06100 (2022)
- A Gaseous Argon-Based Near Detector to Enhance the Physics Capabilities of DUNE, arXiv:2203.06281 (2022)
- Measurement of the double-differential muon-neutrino charged-current inclusive cross section in the NOvA near detector, arXiv:2109.12220 (2023)
- Design, construction and operation of the ProtoDUNE-SP Liquid Argon TPC, arXiv:2108.01902 (2022)
- Searching for solar KDAR with DUNE, arXiv:2107.09109 (2021)
- Extended search for supernovalike neutrinos in NOvA coincident with LIGO/Virgo detections, arXiv:2106.06035 (2021)
- Search for Active-Sterile Antineutrino Mixing Using Neutral-Current Interactions with the NOvA Experiment, arXiv:2106.04673 (2021)
- Seasonal variation of multiple-muon cosmic ray air showers observed in the NOvA detector on the surface, arXiv:2105.03848 (2021)
- Deep Underground Neutrino Experiment (DUNE) Near Detector Conceptual Design Report, arXiv:2103.13910 (2021)
- Experiment Simulation Configurations Approximating DUNE TDR, arXiv:2103.04797 (2021)
- Search for slow magnetic monopoles with the NOvA detector on the surface, arXiv:2009.04867

- (2021)
- Prospects for beyond the Standard Model physics searches at the Deep Underground Neutrino Experiment, arXiv:2008.12769 (2021)
- Supernova neutrino burst detection with the Deep Underground Neutrino Experiment, arXiv:2008.06647 (2021)
- First results on ProtoDUNE-SP liquid argon time projection chamber performance from a beam test at the CERN Neutrino Platform, arXiv:2007.06722 (2020)
- Neutrino interaction classification with a convolutional neural network in the DUNE far detector, arXiv:2006.15052 (2020)
- Long-baseline neutrino oscillation physics potential of the DUNE experiment, arXiv:2006.16043 (2020)
- Adjusting neutrino interaction models and evaluating uncertainties using NOvA near detector data, arXiv:2006.08727 (2020)
- Supernova neutrino detection in NOvA, arXiv:2005.07155 (2020)
- Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume III: DUNE Far Detector Technical Coordination, arXiv:2002.03008 (2020)
- Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume I Introduction to DUNE, arXiv:2002.02967 (2020)
- Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume II: DUNE Physics, arXiv:2002.03005 (2020)
- Deep Underground Neutrino Experiment (DUNE), Far Detector Technical Design Report, Volume IV: Far Detector Single-phase Technology, arXiv:2002.03010 (2020)
- Search for multimessenger signals in NOvA coincident with LIGO/Virgo detections, arXiv:2001.07240 (2020)
- First Measurement of Neutrino Oscillation Parameters using Neutrinos and Antineutrinos by NOvA, arXiv:1906.04907 (2019)
- Observation of seasonal variation of atmospheric multiple-muon events in the NOvA Near Detector, arXiv:1904.12975 (2019)
- Measurement of neutrino-induced neutral-current coherent π^0 production in the NOvA near detector, arXiv:1902.00558 (2020)
- The DUNE Far Detector Interim Design Report Volume 1: Physics, Technology and Strategies, arXiv:1807.10334 (2018)
- The DUNE Far Detector Interim Design Report, Volume 2: Single-Phase Module, arXiv:1807.10327 (2018)
- The DUNE Far Detector Interim Design Report, Volume 3: Dual-Phase Module, arXiv:1807.10340 (2018)
- New constraints on oscillation parameters from ν_e appearance and ν_μ disappearance in the NOvA experiment, arXiv:1806.00096 (2018)
- Results and Prospects from NOvA (2018)
- Event Reconstruction in the NOvA Experiment, arXiv:1710.03772 (2017)
- Sterile Neutrino Search in the NOvA Far Detector, arXiv:1710.01280 (2017)
- NOvA Short-Baseline Tau Neutrino Appearance Search, arXiv:1710.00295 (2017)
- A search for sterile neutrinos at the NOvA Far Detector (2017)
- The Single-Phase ProtoDUNE Technical Design Report, arXiv:1706.07081 (2017)
- Search for active-sterile neutrino mixing using neutral-current interactions in NOvA, arXiv:1706.04592 (2017)
- Constraints on Oscillation Parameters from ν_e Appearance and ν_μ Disappearance in NOvA, arXiv:1703.03328 (2017)
- Measurement of the neutrino mixing angle θ_{23} in NOvA, arXiv:1701.05891 (2017)
- Searches for Sterile Neutrinos with NOvA (2016)
- Physics potential of a long-baseline neutrino oscillation experiment using a J-PARC neutrino beam and Hyper-Kamiokande, arXiv:1502.05199 (2015)
- A Long Baseline Neutrino Oscillation Experiment Using J-PARC Neutrino Beam and Hyper-Kamiokande, arXiv:1412.4673 (2014)

- Expression of Interest: The Atmospheric Neutrino Neutron Interaction Experiment (ANNIE), arXiv:1402.6411 (2014)
- Hyper-Kamiokande Physics Opportunities, arXiv:1309.0184 (2013)
- The Electromagnetic Calorimeter for the T2K Near Detector ND280, arXiv:1308.3445 (2013)
- The Long-Baseline Neutrino Experiment: Exploring Fundamental Symmetries of the Universe, arXiv:1307.7335 (2013)
- Scientific Opportunities with the Long-Baseline Neutrino Experiment (2013)
- CHerenkov detectors In mine PitS (CHIPS) Letter of Intent to FNAL, arXiv:1307.5918 (2013)
- Cosmic Ray Backgrounds in an LBNE far detector on the Surface, arXiv:1206.3561 (2012)
- Fundamental Physics at the Intensity Frontier, arXiv:1205.2671 (2012)
- The Long Baseline Neutrino Experiment (LBNE) Water Cherenkov Detector (WCD) Conceptual Design Report (CDR), arXiv:1204.2295 (2012)
- First Muon-Neutrino Disappearance Study with an Off-Axis Beam, arXiv:1201.1386 (2012)
- Understanding and Calibrating the Calorimeter for the T2K Near Detector (2012)
- Measurements of the T2K neutrino beam properties using the INGRID on-axis near detector, arXiv:1111.3119 (2012)
- NOvA: Present and Future, arXiv:1110.0112 (2011)
- The 2010 Interim Report of the Long-Baseline Neutrino Experiment Collaboration Physics Working Groups, arXiv:1110.6249 (2011)
- The T2K Experiment, arXiv:1106.1238 (2011)
- Indication of Electron Neutrino Appearance from an Accelerator-produced Off-axis Muon Neutrino Beam, arXiv:1106.2822 (2011)
- Understanding and calibrating the DS-ECal for T2K's ND280 (2011)
- The T2K experiment and downstream electromagnetic calorimeter (2008)
- Radiation hardness studies in a CCD with high-speed column parallel readout, arXiv:0711.3561 (2008)
- Simulations of the Temperature Dependence of the Charge Transfer Inefficiency in a High-Speed CCD (2007)
- Progress with vertex detector sensors for the International Linear Collider (2007)
- Radiation Hardness of CCD Vertex Detectors for the ILC, arXiv:physics/0611272 (2007)